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Smart Cities (Urban Simulation, Big Data)

CityFeel - micro climate monitoring for climate mitigation and urban design

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Abstract

While a significant part of the population is concentrated in urban areas, the influence of cityscape parameters on human heat stress remain poorly understood. Yet we agree to develop urban spaces (street, square, district ...) in a way to provide best possible quality of life. In order to do so, quantitative and qualitative references are required. To fill this gap the HES-SO† - hepia/leea‡ has developed an innovative portable monitoring system that can be easily deployed in various outdoor and indoor environments. The monitoring equipment is embedded into a backpack that is carried during 'climatic urban walks' that can be reproduced at different times of the day or seasons so to yield a detailed and dynamic description of the climatic context of a portion of the city from the pedestrian point of view.

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Keywords: climate mitigation; micro climate monitoring; urban design; open data

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1. Main text

1.1. Introduction

75% of the European population is concentrated in rapidly growing urban areas. While adaptation to climate change in cities is a major challenge, "well-being" in terms of hygrothermal comfort and air quality of residents and pedestrians and the contributing effects are still poorly understood. Ongoing urban developments (streets, squares, plazas, parks, crossroads, and neighborhoods) claim to offer the best possible experience to their users (inhabitants, pedestrians). Yet, development policies are often based on subjective knowledge and lack quantitative and qualitative reference data.

Hence, climate warming and urban heat island effect have direct impact on inhabitants' health (specially the elderly and weak), on pollution levels, on energy consumption (e.g. air conditioning), on biodiversity, etc… Due to these issues, growing attention is being paid to the management of urban spaces of all kinds. Design (ergonomics, aesthetics) is not enough; climate resilient planning will be a challenge for the next decade and will help to mitigate adverse effects of climate change. Local authorities will adapt urban planning in ways that take into account climatic and sanitary constraints, meaning that urban planners and local governments require reliable reference data to make best choices with a limited budget and time.

CityFeel aims to provide a toolbox to urban planners for climate-adapted city development. This toolbox comprises a methodology, physical tools and open data that are made available to planners and the scientific community. In-depth knowledge of the dynamics of "comfort and air quality at a pedestrian level" will have many purposes, including but not limited to urban surfaces, vegetation, traffic control, health monitoring, etc.

1.2. State of the art

Unlike reference climate stations (physical parameters) and air quality stations (chemical parameters) that are commonly used for conventional meteorological purposes or the monitoring of pollution levels, *CityFeel* measures the human perception of the immediate environment in terms of response to hygrothermics and air quality. This is why the protocol must be tied to the human metabolism and measurements must take place at pedestrian level.

Most commonly, when climate or air quality data are available for a given city, these are derived from static measurement stations. Their aim is to qualify the urban climate according to conventional classifications such as city center, suburbs, countryside. For example Geneva has three WMO grade measuring stations, (maintained by the University of Geneva and by hepia / climacity) in addition to the historical station of the Swiss Office of Meteorology at Cointrin (Geneva airport). The size and complexity of these conventional stations do not allow for extensive coverage over a given territory or even a district. Furthermore, measurements from conventional weather stations do not describe what is "felt" at pedestrian level.

Other methods consist of making use of numerical simulation models; whilst the complexity of these algorithms can be managed with increase of computing power, limitations are mainly due to the capability of physical model input description and limited resolution of the city model, as well as the scarcity of reliable data for model calibration. These models tend to have limited resolution and reliability at pedestrian level, where the feedback is most critical and also is most complex to simulate.

Drive- or walkthroughs with sensors attached to vehicles have been demonstrated over the past decades and have produced parameter profiles trough many city sections. The observation that for example 'green' or 'blue' do not always have measurable effect shows that there still is a lack of knowledge on how these indicators should be interpreted into robust 'climatic bricks' for urban developers and planners.

Due to the previous points we have not yet found a city with a network of long-term quality climate measurements at pedestrian level with a high density of measurement points and who's monitoring considers human metabolism.

1.3. Methodology

The first step was to define an investigation methodology and the qualitative criteria based on the multiple senses that affect the complex and multisensory perception of pedestrians while roaming through the city (e.g. the human body is sensitive to radiant temperature from surrounding surfaces, to air turbulence, to noise, to odors, to visual and acoustic perception, and so on…).

Then, knowing that the deployment of a fixed monitoring network with as many data collection points as necessary would be too restrictive, too expensive and difficult to achieve, we changed the paradigm by developing a portable and affordable measuring device in the form of a backpack. In this way, the sensors are attached to a pedestrian operator and move along a free or predefined path through the city. This provides great flexibility of use, anywhere and at any given time. The data is collected at a rate of 1 Hz and is made available for display and analysis by the means of a dedicated graphical interface or more traditional analysis tools.

Hemispherical and ground images complete the sensor array. They are used for analysis, such as sky view factor or vegetation index, but also allow for 'reverse engineering' of acquired data: in a way they give eyes to the sensors and allow better understanding of cause and effect. Data collected from the sensors feed a human body heat balance model to evaluate physiological equivalent temperature (PET), which is defined as equivalent temperature having comparable heat sensation in controlled environmental conditions without radiation, wind, etc. PET is easily understandable and is closest to the actual heat sensation.



Fig. 1. (left) operator during a climate walk, carrying the microclimate-meter, (right) sample of data displayed in the ad-hoc viewer

2. Description

2.1. Investigation and measurement equipment

The specifically designed 'microclimate-meter', further referred to as *uCM*, is the core part of *CityFeel*. The goal is to deploy an array of environmental sensors within a mobile and light-weight system.

Thus, beyond the conventional parameters (ambient temperature and humidity), the *uCM* also logs multi-directional radiation temperatures, vertical temperature profiles at pedestrian level, thermal turbulence, wind speed, direction and turbulence, noise level and records the soundtrack and hemispherical images (sky and ground camera). A GPS receiver and an inertial reference platform make it possible to locate the measured data in time and in space. In the latest version we added additional sensors to measure air quality such as CO2, NOx, Ozone and micro particles.

Table 1. Table of physical sensors:

Sensor	Resolution	Absolute error
Ambient temperature	0.1°C	0.2°C
High speed temperature	0.1°C	0.4°C
Relative humidity	0.1%	1%
Radiant temperature (left, right … up, down)	0.1°C	0.5°C
Solar radiation	1W/m2	1.5%
Wind speed	0.01m/s	1%
Sound pressure	1dB	3%
Position (GPS)	0.1m	<10m

2.2. The uCM on-board equipment

The sensor array is connected to a dedicated logging equipment consisting of a Campbell Scientific CR-6 data logger, an ATmega328 based micro-controller board and a nano-PC for image handling and on-the-fly data quality assessment and synchronization. Ongoing development includes a real-time graphical operator interface.

The approximate data flux is 7MB/second. The autonomy is about three hours of continuous sensor streaming, limited only to LiPo battery and memory stick capacity and capacity of the operator to carry the 5kg equipment.

2.3. Data processing and results

During the development process, calibration test walks followed a fixed reference sequence, characterized by a sufficiently contrasting "urban scape" (avenue, square, park, street). At present, the operator simply puts on the *uCM* and starts monitoring according to specific interests.

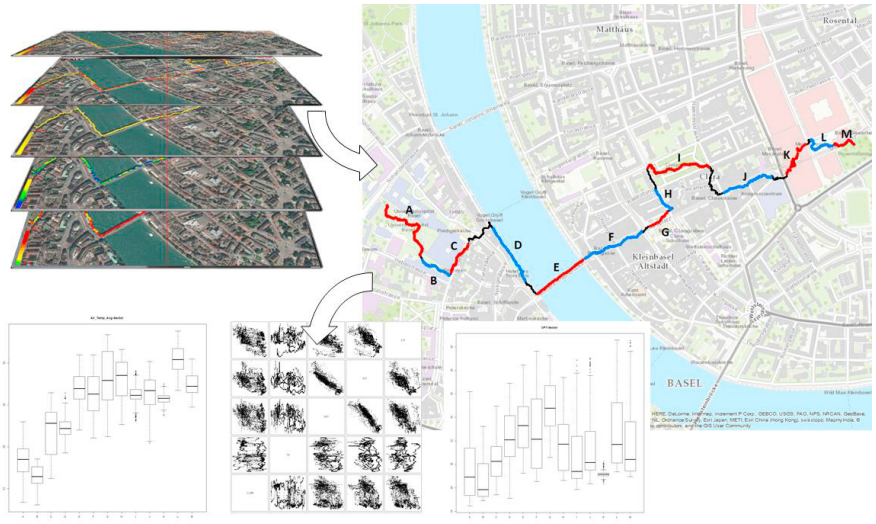


Fig. 2. Overview of the process of data analysis: crude data visualization, sectorization, statistical analysis

Once the campaign is finished, the operator transfers the collected data to a desktop PC where a script aggregates all collected data, converts it to a suitable format for post-treatment and calculates derived indicators related to outdoor human heat stress, such as mean radiant and operative temperatures, PET (Physiologically Equivalent Temperature), UTCI (Universal Thermal Climate Index), HUMIDEX (Humidity index), WBGT (Wet Bulb Globe Temperature), etc.

A Java™ program allows for first observation and interpretation of the collected data by the means of a dedicated graphical user interface. In depth analysis is done by traditional environments, such as e.g. ArcGIS mapping and

analytics software, Mathlab®, Statistica® or R, typically used to sectorise, analyze and visualize data. Machine learning algorithms take care of related image analysis, such as sky view factor (SVF) or vegetation indexes.

2.4. Case-studies

The interaction of city and climate is a very complex process and is determined by many influencing factors. Added to this are weather conditions, time of day, use (differences between works and holidays) and much more. Never less, assuming that variations over a short period of time are small, a close look into measurements data give clues to comprehension of the urban metabolism.

How efficient are climate strategies in the real-world and what are the best options can be identified by the observation of comparative solutions at different periods and times of the day. For example evaporation from water effectively acts as a cooling strategy, as can be seen on the illustration hereafter.

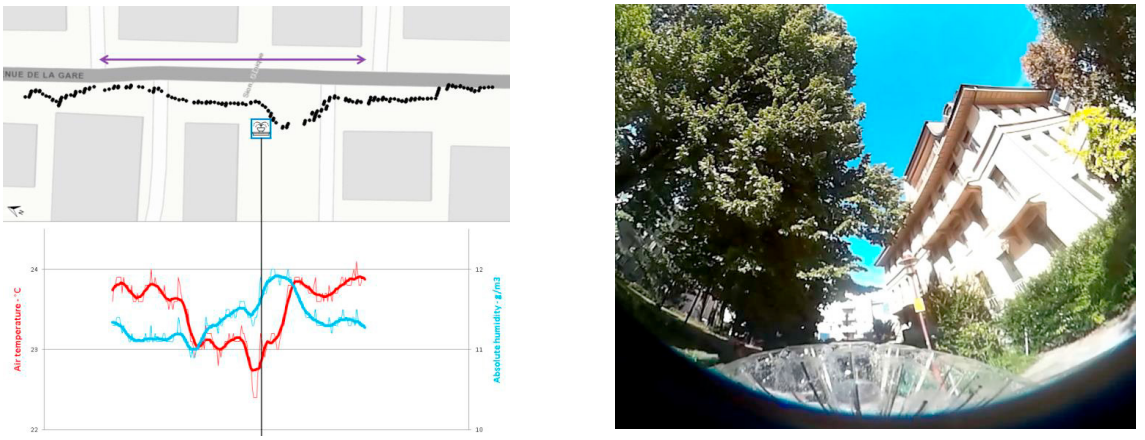


Fig. 3. Example effect of water as urban climatic design feature decreasing temperature whilst increasing air water content.

Climate walks are also useful to identify places where heat stress and pollution are least. This knowledge can be used to inform the population and to identify local adaptation and mitigation strategies. Charts showing parameters such as PET, UTCI, humidity, wind, etc… show a strong correlation between urban morphology and parameters affecting human outdoor comfort and health.

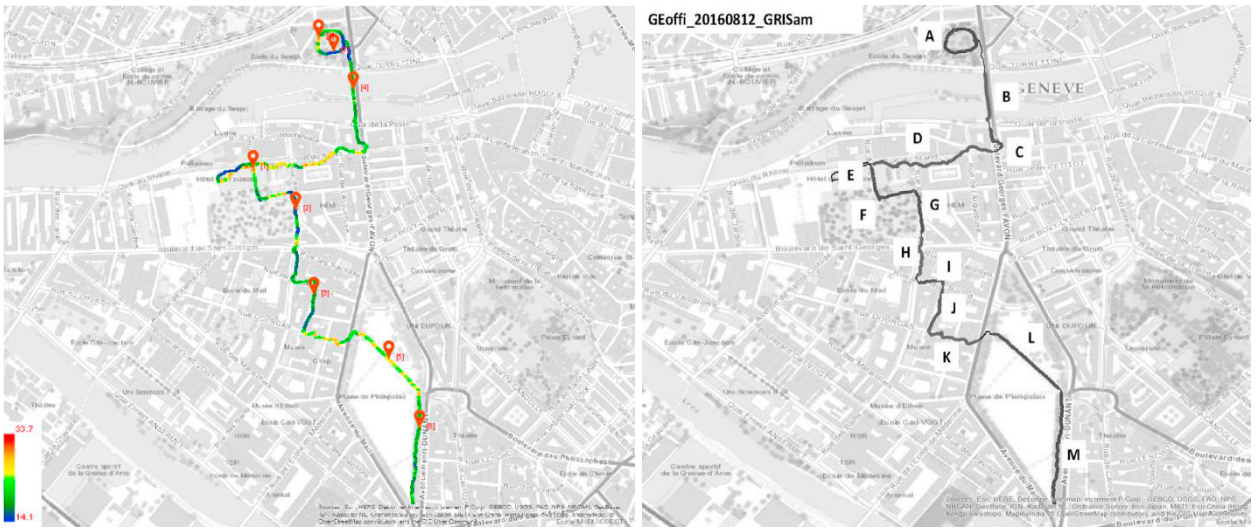


Fig. 4. PET profile of a climate walk in Geneva center indicating cold- and hot spots on a summer morning and sectors (right)

It has been shown that already a single climate walk can be meaningful. The following comparison of trends for a series of thirteen sectors (A to M, see figure 4 right side) on different days is very good. This means that even though the absolute temperature may not be known in advance, the data of a single walk can be extrapolated to predict thermal behavior of the city and give objective indications to questions such as: what are the best places to rest, which are the most comfortable and healthiest ways to travel from one place to another depending on the time of day, and so on.

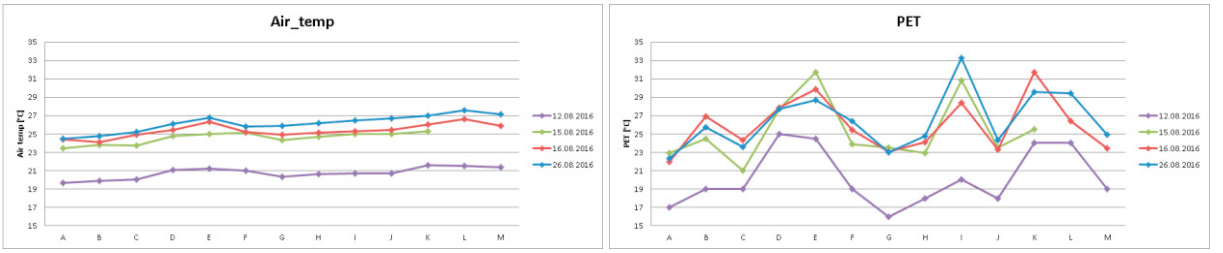


Fig. 5. Comparison of temperature and PET profiles for different days; sectors refer to figure 4

2.5. Discussion and outlook

The *uCM* monitoring and logging backpack has contributed in research projects related to effects of vegetation and water in the city and urban morphology. Numerous climatic walkthroughs have been completed over the last year revealing interesting insight into climatic metabolism of cities.

The *uCM* is readily available for research in the field of city climate, city- and soundscapes (well-being) and air quality (health). On the basis of 'crowdsharing' the *uCM* is made available for free to public and academic partners who are interested in investigating, contributing and sharing studies through an open-data structure containing monitored situations relating to climatic urban design of public spaces and air quality. The collected data will be shared amongst the scientific community by the means of an open data structure. This knowledge will be useful in applications such as urban planning, choice of urban surfaces, design of the "green city" and uses such as traffic control, pollution prediction, health assessment as well as general urban management.

The *uCM* is now in version 2.0 and ongoing hard- and software development aim to make the *uCM* ever more immersive, easier and intuitive to use.

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